

In the 1930s, however, precise measurements by astronomers of the nebulae – small whitish clouds between the stars – showed that the Galaxy itself is a speck of dust in a huge cloud of galaxies, which extends as far as the eye can see using even our most powerful telescopes. The world has now become a uniform and boundless expanse.

The illustration below is not a drawing; it's a photograph taken by the Hubble telescope in orbit, showing a deeper image of the sky than any seen previously with the most powerful of our telescopes: seen with the naked eye it would be a minute piece of extremely black sky. Through the Hubble telescope a dusting of vastly distant dots appears. Each black dot in the image is a galaxy containing a hundred billion suns similar to ours. In the past few years it has been observed that the majority of these suns are orbited by planets. There are therefore in the universe thousands of billions of billions of billions of planets such as Earth. And in every direction in which we look, this is what appears: